



**GOVERNMENT OF KARNATAKA
KARNATAKA SCHOOL EXAMINATION AND ASSESSMENT BOARD**

Model Question Paper - 2

II P.U.C: MATHEMATICS (35): 2026-27

Time: 3 hours

Max. Marks: 80

Instructions:

- 1) The question paper has five parts namely A, B, C, D and E. Answer all the parts.
- 2) PART A has 15 MCQ's, 5 Fill in the blanks of 1 mark each.
- 3) Use the graph sheet for question on linear programming in PART E.
- 4) For questions having figure/graph, alternate questions are given at the end of question paper in separate section for visually challenged students.

PART -A

I. Answer All The Multiple Choice Questions:

15 x 1 = 15

1. If a relation R on the set $\{1, 2, 3\}$ be defined by $R = \{(1, 1), (2, 2)\}$, then R is
 (A) symmetric but not transitive (B) transitive but not symmetric
 (C) symmetric and transitive. (D) neither symmetric nor transitive.
2. Match Column I with Column II

Column I	Column II
a) Domain of $\sec^{-1} x$	i) $R - (-1, 1)$
b) Range of $\operatorname{cosec}^{-1} x$	ii) $(0, \pi)$
c) Range of $\cot^{-1} x$	iii) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

Choose the correct answer from the options given below:

- (A) a-i, b-ii, c-iii (B) a-iii, b-ii, c-i (C) a-i, b-iii, c-ii (D) a-iii, b-i, c-ii
3. The matrix $A = \begin{bmatrix} 0 & 2b & -2 \\ 3 & 1 & 3 \\ 3a & 3 & -1 \end{bmatrix}$ is a symmetric matrix. Then the value of a and b respectively are:
 (A) $-\frac{2}{3}, \frac{3}{2}$ (B) $-\frac{1}{2}, \frac{1}{2}$ (C) -2, 2 (D) $\frac{3}{2}, \frac{1}{2}$
4. A and are invertible matrices of the same order such that $|AB| = 8$ if $|A| = 2$ then $|B|$ is
 (A) 4 (B) 16 (C) 1/4 (D) 1/16
5. The function $f(x) = [x]$, where $[.]$ is greatest integer function is _____
 (A) discontinuous only for integral x (B) a constant function
 (C) continuous for all x (D) derivable for all x
6. The derivative of $e^{\log_e x}$ with respect to x is
 (A) $e^{\log_e x}$ (B) $\frac{1}{x}$ (C) 1 (D) $\frac{e^{\log_e x^2}}{x}$

7. The total revenue in rupees received from the sale of x units of a product is $R(x) = 3x^2 + 36x + 5$, then the marginal revenue when $x = 15$ is Rupees
 (A) 116 (B) 96 (C) 90 (D) 126
8. The absolute maximum value of the function f given by $f(x) = x^3$, $x \in [-2, 2]$ is =.....
 (A) -2 (B) 2 (C) 0 (D) 8

9. $\int \frac{2-3\sin x}{\cos^2 x} dx =$

- (A) $2\tan x - 3\sec x + c$ (B) $2\tan x + 3\sec x + c$
 (C) $2\tan x - 2\sec x + c$ (D) $2\sec x - 3\tan x + c$

10. $\int_1^{\sqrt{3}} \left(\frac{1}{1+x^2} \right) dx =$

- (A) $\frac{\pi}{3}$ (B) $\frac{2\pi}{3}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{12}$

11. **Statement 1:** If either $|\vec{a}| = 0$ or $|\vec{b}| = 0$ then $\vec{a} \cdot \vec{b} = 0$

Statement 2: If $\vec{a} \times \vec{b} = \vec{0}$, then $\vec{a} \perp \vec{b}$.

- (A) Statement 1 is true and Statement 2 is false
 (B) Statement 1 is false and Statement 2 is true
 (C) Statement 1 is true and Statement 2 is true
 (D) Statement 1 is false and Statement 2 is false

12. If $|\vec{a} \cdot \vec{b}| = -|\vec{a}||\vec{b}|$ then the angle between $\vec{a}$ & $\vec{b}$ is

- (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) π (D) $\frac{\pi}{3}$

13. The direction ratios of x -axis.

- (A) 0, k, 0 (B) 0, 0, k (C) k, 0, 0 (D) k, k, k

14. If $P(A) = \frac{1}{2}$ and $P(B) = 0$ then find $P(B|A)$ is

- (A) $\frac{1}{2}$ (B) 0 (C) 1 (D) not defined

15. Two balls are drawn at random with replacement from a box containing 10 black and 8 red balls, then the probability that both are red

- (A) $\frac{25}{81}$ (B) $\frac{16}{81}$ (C) $\frac{20}{81}$ (D) $\frac{28}{153}$

II. Fill in the blanks by choosing the appropriate answer from those given in the

bracket (0, -1, 2, 3, 5, 10)

5 x 1 = 5

16. $\frac{d}{dx}(3x^x)$ at $x=1$ is = ____

17. Minimum value of $f(x) = |x + 2| - 1$ is ____

18. Find the number of arbitrary constants in the particular solution of a differential equation of third order is ____

19. The scalar product of $\lambda i + j - 3k$ and $3i - 4j + 7k$ is -10, then the value of λ is ____

20. If $P(A) = \frac{3}{5}$, $P(B) = \frac{5}{10}$ and A and B are independent events then $P(A \cap B) = \frac{3}{k}$, then $k =$ ____

PART - B**III. Answer Any Six Questions****6 x 2 = 12**

21. Prove that $\sin^{-1}\left(2x\sqrt{1-x^2}\right) = 2\sin^{-1}x$, $-\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$
22. Find the equation of the line joining the points (3, 1) and (9, 3) using determinants
23. Find $\frac{dy}{dx}$, if $y + \sin y = \cos x$
24. Find the rate of change of the area of a circle with respect to its radius r at $r = 6$ cm.
25. Find $\int \sin 2x \cos 3x dx$
26. Find the general solution of the differential equation $\frac{dy}{dx} = \frac{x+1}{2-y}$, ($y \neq 2$)
27. Find the projection of the vector $\vec{a} = 2\vec{i} + 3\vec{j} + 2\vec{k}$ on the vector $\vec{b} = \vec{i} + 2\vec{j} + \vec{k}$.
28. Find the vector and Cartesian equation of the line passing through the point (5, 2, -4) and parallel to the vector $3\vec{i} + 2\vec{j} - 8\vec{k}$
29. A die is tossed thrice. Find the probability of getting an odd number at least once

PART - C**IV. Answer Any Six Questions****6 x 3 = 18**

30. Show that the relation R on the set $A = \{x \in \mathbb{Z} : 0 \leq x \leq 12\}$, given by
 $R = \{(a, b) : |a - b| \text{ is a multiple of } 4\}$ is an equivalence relation
31. Prove that $\cos^{-1} \frac{4}{5} + \cos^{-1} \frac{12}{13} = \cos^{-1} \frac{33}{65}$
32. Prove that for any square matrix A with real number entries, $A + A'$ is a symmetric matrix and $A - A'$ is a skew symmetric matrix.
33. Find $\frac{dy}{dx}$, if $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$.
34. Find the intervals in which the function f given by $f(x) = 2x^3 - 3x^2 - 36x + 7$ is strictly decreasing
35. Find $\int x \tan^{-1} x dx$
36. Find the unit vector perpendicular to each of the vectors $(\vec{a} + \vec{b})$ & $(\vec{a} - \vec{b})$, where
 $\vec{a} = 3\vec{i} + 2\vec{j} + 2\vec{k}$ & $\vec{b} = \vec{i} + 2\vec{j} - 2\vec{k}$.
37. Find the distance between the lines
 $\vec{r} = \vec{i} + 2\vec{j} - 4\vec{k} + \lambda(2\vec{i} + 3\vec{j} + 6\vec{k})$ & $\vec{r} = 3\vec{i} + 3\vec{j} - 5\vec{k} + \mu(2\vec{i} + 3\vec{j} + 6\vec{k})$
38. Of the students in a college, it is known that 60% reside in hostel and 40% are day scholars (not residing in hostel). Previous year results report that 30% of all students who reside in hostel attain A grade and 20% of day scholar attain A grade in their annual examination. At the end of the year, one student is chosen at random from the college and he has an A grade, what is the probability that the student is a hostile?

PART – D

V. Answer Any Four Questions

4 x 5 = 20

39. Consider $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 4x + 3$. Show that f is invertible. Find the inverse.

40. If $A = \begin{bmatrix} 1 & 2 & -3 \\ 5 & 0 & 2 \\ 1 & -1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -1 & 2 \\ 4 & 2 & 5 \\ 2 & 0 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 4 & 1 & 2 \\ 0 & 3 & 2 \\ 1 & -2 & 3 \end{bmatrix}$

Then compute $(A + B)$ and $(B - C)$. Also, verify that $A + (B - C) = (A + B) - C$

41. Solve the system of equations $x + y + z = 6$, $y + 3z = 11$ and $x - 2y + z = 0$ by matrix method.

42. If $y = \sin^{-1} x$, then prove that $(1 - x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} = 0$.

43. Find the integral of $\frac{1}{x^2 - a^2}$ with respect to x and evaluate $\int \frac{dx}{x^2 - 16}$

44. Using the method of integration, find the area enclosed by the circle $x^2 + y^2 = a^2$

45. Find the general solution of the differential equation $\frac{dy}{dx} + y \cot x = 4x \operatorname{cosec} x (x \neq 0)$.

PART – E

VI. Answer The Following Questions

46. Prove that $\int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx & \text{if } f(x) \text{ is even} \\ 0 & \text{if } f(x) \text{ is odd} \end{cases}$ hence evaluate $\int_{-\pi/2}^{\pi/2} (x^3 + x \cos x + \tan^5 x) dx$. **6M**

OR

Solve the following graphically, Minimize $Z = 200x + 500y$, subject to the constraints $x + 2y \geq 10$, $3x + 4y \leq 24$, $x \geq 0$, $y \geq 0$.

47. If $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ satisfies the equation $A^2 - 4A + I = O$, then find the inverse of A using this equation, where I is the identity matrix of order 2 and O is the zero matrix of order 2 **4M**

OR

Determine the value of k , if $f(x) = \begin{cases} \frac{k \cos x}{\pi - 2x}, & \text{if } x \neq \frac{\pi}{2} \\ 3, & \text{if } x = \frac{\pi}{2} \end{cases}$ is continuous at $x = \frac{\pi}{2}$.